

GeoCalib + Depth Pro

Real Estate Room Distance Measurement

Pipeline Validation & Accuracy Evaluation

1. Research Background

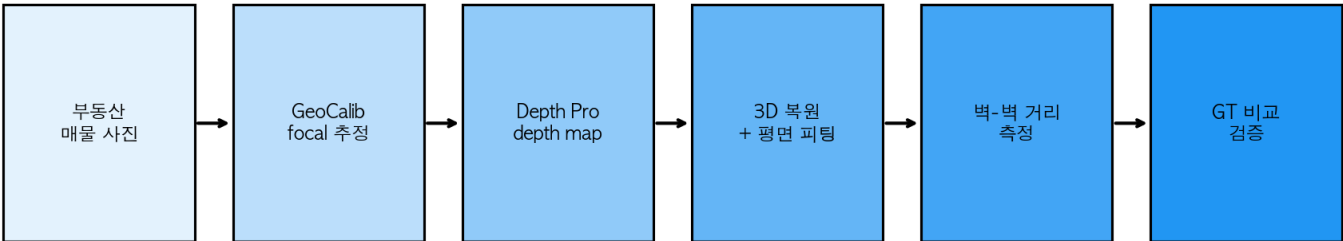
Real estate listing photos use wide-angle lenses (FOV 90-130 degrees), making rooms appear larger. Platforms strip EXIF metadata, hiding lens information from consumers.

Goal: Automatically measure wall-to-wall distance from a single real estate photo using GeoCalib (focal estimation, ECCV 2024) + Depth Pro (metric depth, ICLR 2025).

Related: Propix (2024) used GeoCalib for distortion detection only, without distance measurement. This study extends Propix by adding actual measurement + GT validation.

2. Pipeline

파이프라인 구조



3. Dataset

Total: 83 photos from 6 apartment complexes

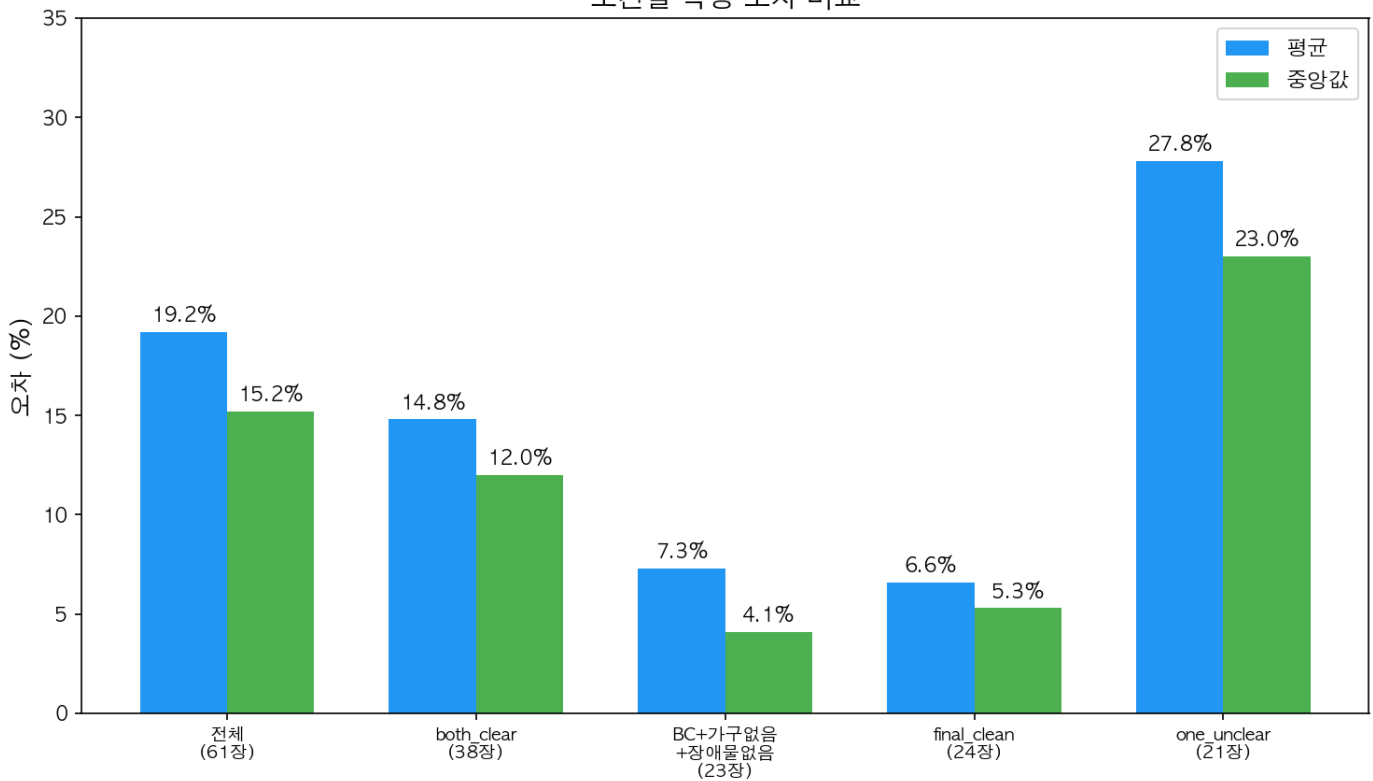
- Original: 63 photos (Olympic Park Foreon, Gaepo Xi, Raemian)
- New Helio City: 9 photos
- New Mapo Raemian: 11 photos

GT: Wall-to-wall distance from floor plans (centerline - 160mm wall thickness)

4. Results by Condition

Condition	Count	Mean Err	Median	<10% Rate
All photos	61	19.2%	15.2%	41%
Both walls clear	38	14.8%	12.0%	55%
BC + No furniture + No obstacle	23	7.3%	4.1%	76%
Final clean	24	6.6%	5.3%	79%
One wall unclear	21	27.8%	23.0%	14%

조건별 측정 오차 비교



5. GeoCalib Focal Length Effect

Comparison: A (GeoCalib focal) vs B (Depth Pro self-estimated focal)

Overall: A=19.2% vs B=20.7% (A wins 64%)

Wide-angle (<500px): A=19.0% vs B=21.5% (A wins 69%)

Standard (>=500px): A=19.6% vs B=19.0% (similar)

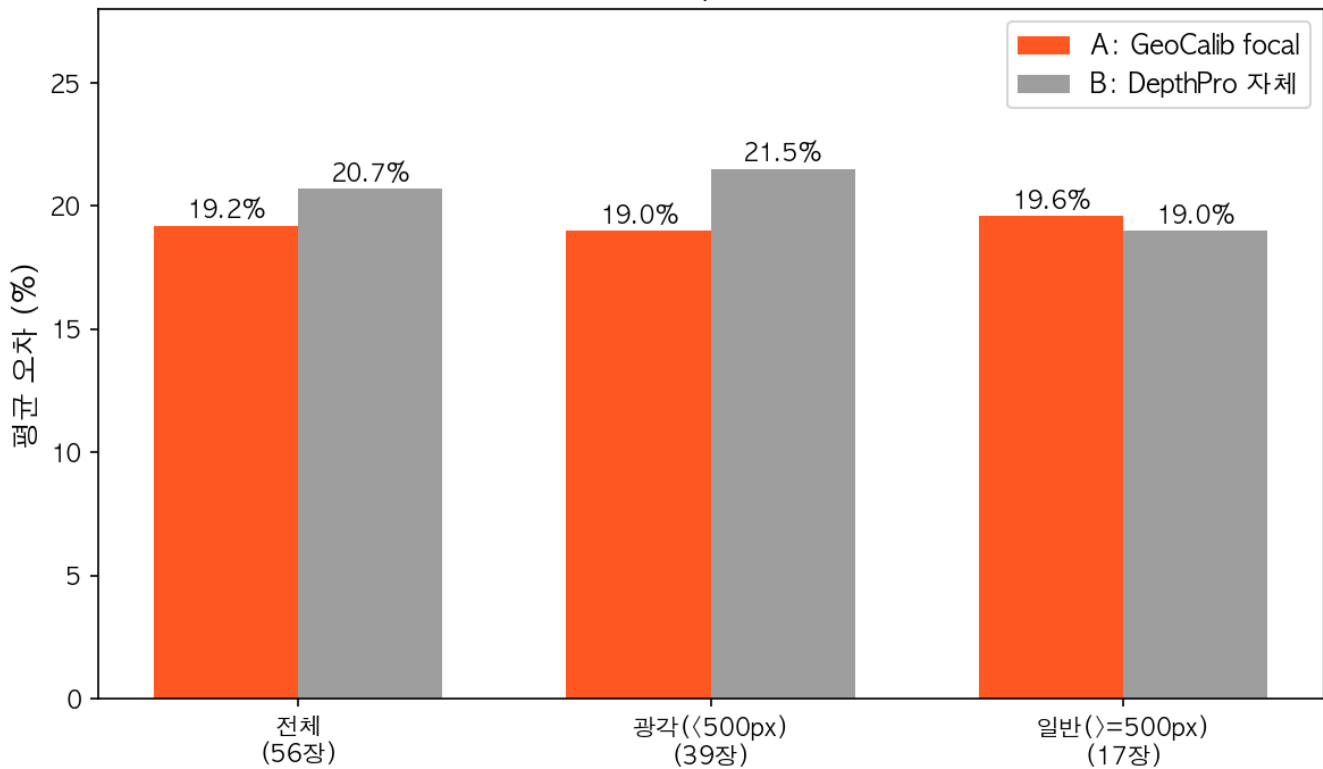
For clean photos only:

Wide-angle: A=11.6% vs B=15.3% (A wins 78%)

Standard: A=3.5% vs B=3.0% (similar)

Conclusion: GeoCalib focal is especially beneficial for wide-angle photos.

GeoCalib focal vs DepthPro 자체 추정 비교



6. Barrel Distortion Effect (ER)

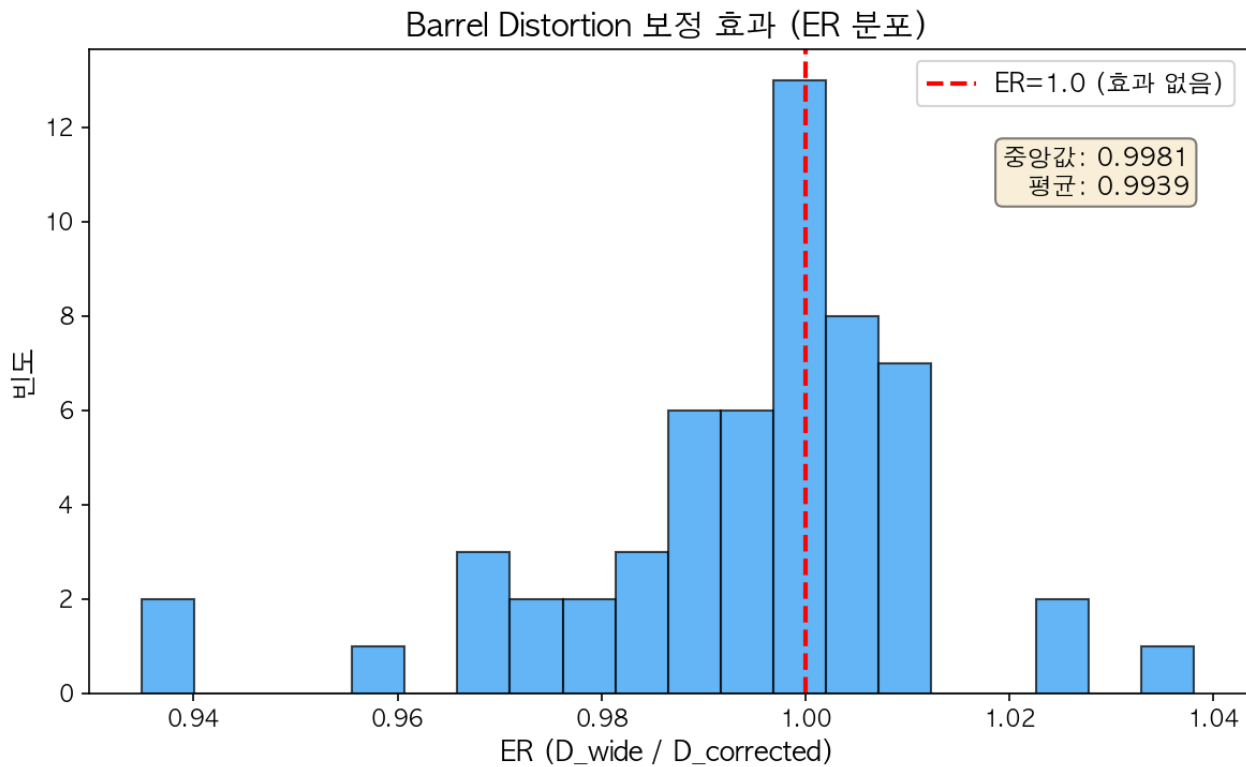
ER = $D_{\text{wide}} / D_{\text{corrected}}$ (original vs GeoCalib corrected)

ER median: 0.998, mean: 0.994, range: 0.935-1.038

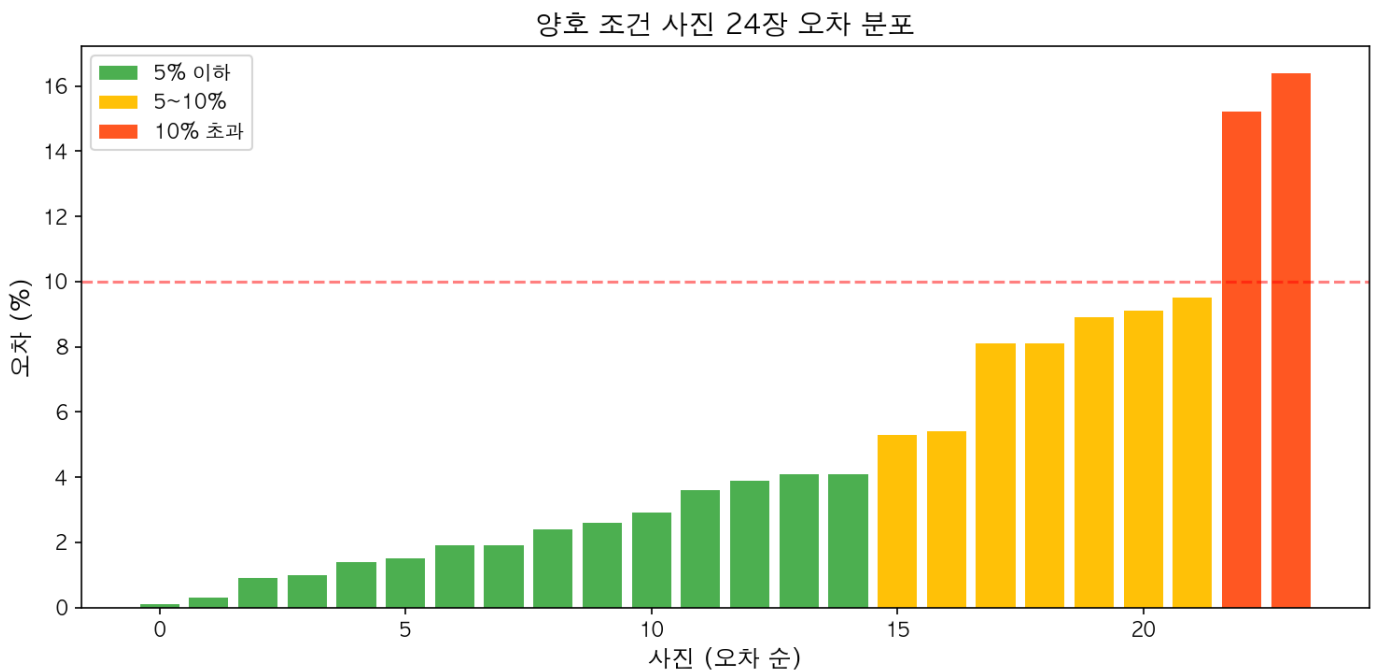
ER ~ 1.0: Barrel distortion correction has NO significant effect on Depth Pro.

This contrasts with Buquet et al. (2021) who found 5% improvement with dewarping.

Modern MMDE models (Depth Pro) appear robust to barrel distortion.



7. Clean Photos Error Distribution (24 photos)

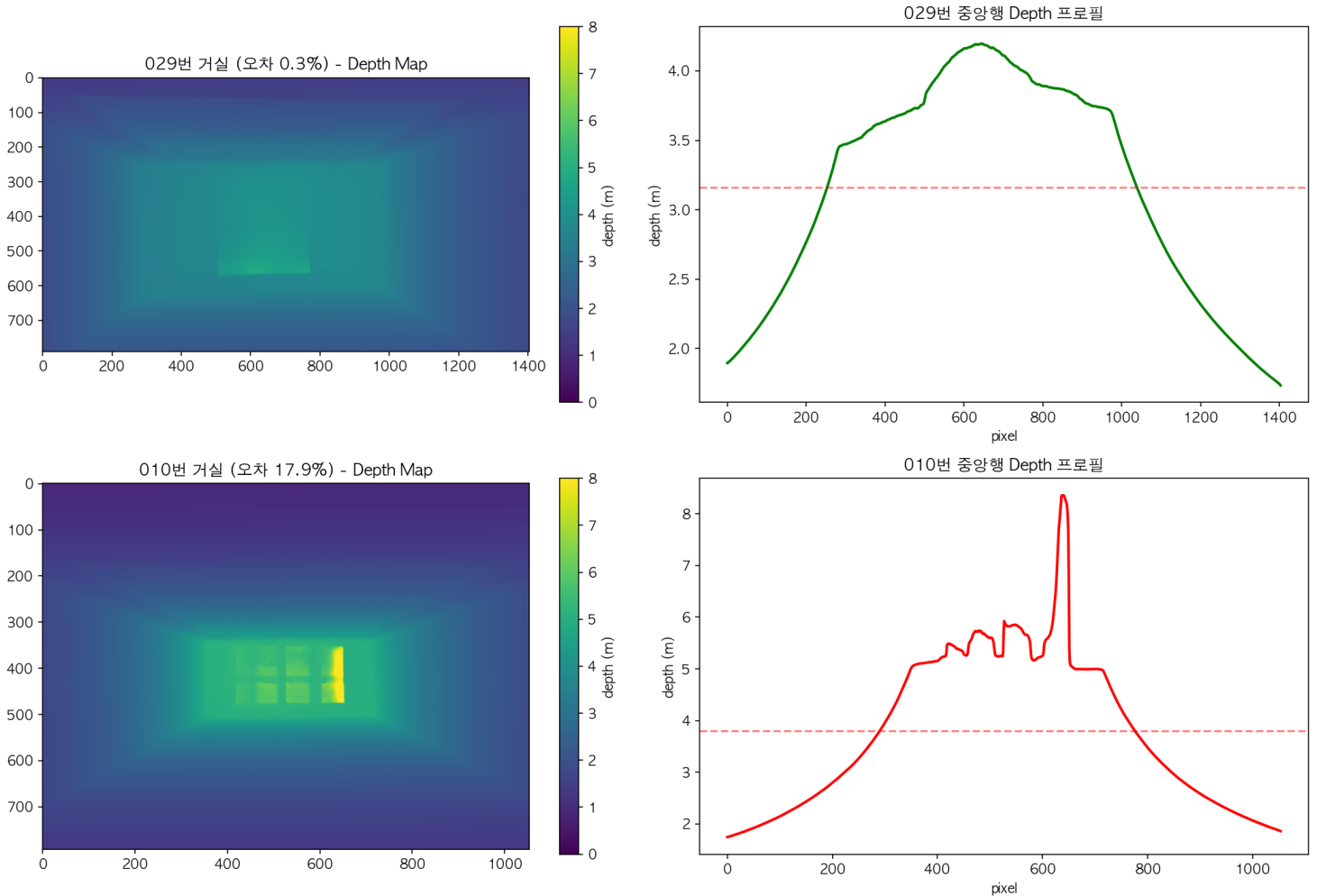


8. Depth Map Comparison

Good photo (#029, 0.3% error) vs Poor photo (#010, 17.9% error)

Both photos have clean conditions, but Depth Pro estimates differently.

Depth Map 비교: 성능 좋은 사진 vs 성능 나쁜 사진



9. Suitable Conditions

Condition	Required	Evidence
Both walls visible	Required	one_unclear: 27.8% vs both_clear: 14.8%
No furniture/obstacles	Required	With obstacles: 24.5% avg error
No balcony glass door as wall	Required	Glass door causes depth leak
Centered/front-facing shot	Recommended	Tilted shots may fail (95.7%)

10. Limitations

1. High variance even under good conditions (0.1%-18.7% for clean photos)
2. Cannot predict which individual photo will be accurate
3. Sample size: 83 total, 24 clean - needs more data
4. Single model (Depth Pro only) - needs Metric3D, UniDepth validation
5. GT uncertainty: wall thickness correction +/-40mm

11. Conclusion

1. GeoCalib + Depth Pro pipeline can measure room width from real estate photos.
2. Under favorable conditions: ~79% of photos achieve <10% error (median 5.3%).
3. GeoCalib focal is effective for wide-angle photos (~4%p improvement).
4. Barrel distortion correction is unnecessary (ER ~ 1.0).
5. Extends Propix (2024) with actual measurement + GT validation.